

ALL WORK, NO PLAY: UMass physicist harnesses the power of Playstations for science

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ARTMOUTH — Guarav Khanna admits with a blush that he didn't let his two young daughters play with the new Sony Playstation 3 his wife gave him on Christmas morning. He was too busy using it to solve the mysteries of the universe.

For real.

Dr. Khanna is a young physics professor at UMass Dartmouth who is among the first in the nation to take advantage of the Playstation 3's little secret: It's not just a toy, it is a supercomputer. That's no exaggeration. And Dr. Khanna has found that Sony fully intends it to be used that way, and soon.

Which brings us back to the universe.

With a small grant from Sony of eight Playstation 3's, Dr. Khanna is running unimaginably complex equations with the intention of predicting the gravity waves that are generated by the super-sized black holes at the centers of galaxies. To make matters more complicated, he is trying to predict the wave patterns that occur when two black holes interact with one another.

"It isn't easy to ask for myself for stuff like this from the NSF (National Science Foundation) or NASA," Dr. Khanna confides. Such agencies are more used to grant applications involving massive supercomputers that are rented to scientists by the hour. Game consoles wouldn't be taken seriously.

"So I took it to Sony directly," he said.

Sony USA was ready for him, more or less.

It said it had a mandate from Tokyo to spread the use of the Playstation 3 with its IBM-produced processor "to benefit humanity and the sciences," he said. They were waiting for the phone to ring, with experimenters proposing new uses.

The thing that is so special about the processor, Dr. Khanna said, is its architecture, which differs from other processors in a major way: The main unit is spared the task of bogging down in huge computations, which instead are farmed out to a half dozen subprocessors.

The result is a computer that runs rings around even the better desktop units now on the market, being the equal of about 25 of them all running at once for his purposes, Dr. Khanna said. That is the power of 25 nodes in the IBM Blue Gene Supercomputer, he said. And since he usually needs eight or more of those nodes to run his programs — the equivalent of 200 to 500 desktops — Dr. Khanna did the simple math and modestly asked Sony back in February for eight Playstation 3's.

It took a few months, but Sony granted his wish and shipped the shiny black Playstations in August. Now they are locked in a steel cage in his office, in two rows of four, with a sign warning that their

game-playing capacities have been permanently disabled, to deter anyone with other nefarious ideas about their use.

They are not much to see, really, and when Dr. Khanna displays their output on his office computer, it's little more than a river of readouts. But Dr. Khanna said that dry runs of programs he has already run on the supercomputers elsewhere have proved the Playstation 3 to be every bit their equal. Now, after the test drives, he is running new calculations on them, with his programs tweaked to take advantage of the unique structure of the new processor.

"The equations are horrendously complicated," he said, which is why traditional processing chips have trouble managing them.

Dr. Khanna is one of only two researchers he knows of — the other is in North Carolina, he said — who have begun to exploit the game system this way. And he said that Sony seems interested in someday marketing a desktop supercomputer based in large part on what scientists can make the Playstation 3 do.

It's not that other game platforms can't perform as well, in case anyone is wondering.

But Dr. Khanna said that Sony is the only manufacturer that has made its source code available, that has made it possible for anyone to reprogram the processor using FORTRAN, the computational language Dr. Khanna is using, Linux, or whatever. Nintendo Wii and XBox 360 are still black boxes, closed off to alternative uses.

By contrast, Sony has intended to open up the Playstation 3 to exploration since well before its introduction last November, he said.

The question now becomes: How many of these would Dr. Khanna like to string together? To which the answer is: as many as possible. But he said he started out with a "modest" request.

"If I had 100 I could be far more accurate," he said. The computations could be compared to the megapixels of resolution on a digital camera, with the finer distinctions revealing small but significant patterns that might otherwise go unobserved, the way a satellite picture of Earth would not reveal without great detail that there are human beings on the surface.

For now, however, Dr. Khanna is happy to be putting his team of Playstations through its paces.

As for his daughters Sarah and Rachel, they got their own Playstation back in March, and they have more conventional intentions for it.

The secrets of the universe are Dad's to figure out.

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